EX NO: 3a	**GENERATION OF PHASE MODULATED (PM) SIGNALS AND CALCULATION OF PHASE DEVIATION USING MATLAB SIMULATION**
DATE:	

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%============================================================
% Experiment: Phase Modulation (PM) and Phase Deviation
%============================================================

clc;
clear;
close all;

%% Parameters
Am = 1;            % Message Amplitude (V)
fm = 100;          % Message Frequency (Hz)

Ac = 5;            % Carrier Amplitude (V)
fc = 1000;         % Carrier Frequency (Hz)

fs = 20 * fc;      % Sampling Frequency (Hz)
kp = pi/2;         % Phase Sensitivity (rad/V)

%% Time Vector
t = 0:1/fs:4/fm;

%% Message Signal
m = Am * sin(2*pi*fm*t);

%% Carrier Signal
c = Ac * cos(2*pi*fc*t);

%% Phase Modulated Signal
pm = Ac * cos(2*pi*fc*t + kp*m);

%% Phase Deviation
phase_dev = kp * Am;

%% PM Modulation Index
mp = phase_dev;

%% Frequency Spectrum
N = length(pm);
f = linspace(-fs/2, fs/2, N);
PM_FFT = abs(fftshift(fft(pm))) / N;

%% Display Results
fprintf('=========== PHASE MODULATION RESULTS ===========\n');
fprintf('Message Amplitude (Am)      = %.2f V\n', Am);
fprintf('Message Frequency (fm)      = %.0f Hz\n', fm);
fprintf('Carrier Amplitude (Ac)      = %.2f V\n', Ac);
fprintf('Carrier Frequency (fc)      = %.0f Hz\n', fc);
fprintf('Sampling Frequency (fs)     = %.0f Hz\n', fs);
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fprintf('Phase Sensitivity (kp)      = %.2f rad/V\n', kp);
fprintf('Phase Deviation             = %.2f radians\n', phase_dev);
fprintf('PM Modulation Index (mp)    = %.2f\n', mp);

%% Plotting
figure('Name','Phase Modulation and Phase Deviation','NumberTitle','off');

% Message Signal
subplot(3,2,1);
plot(t, m, 'b', 'LineWidth', 1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

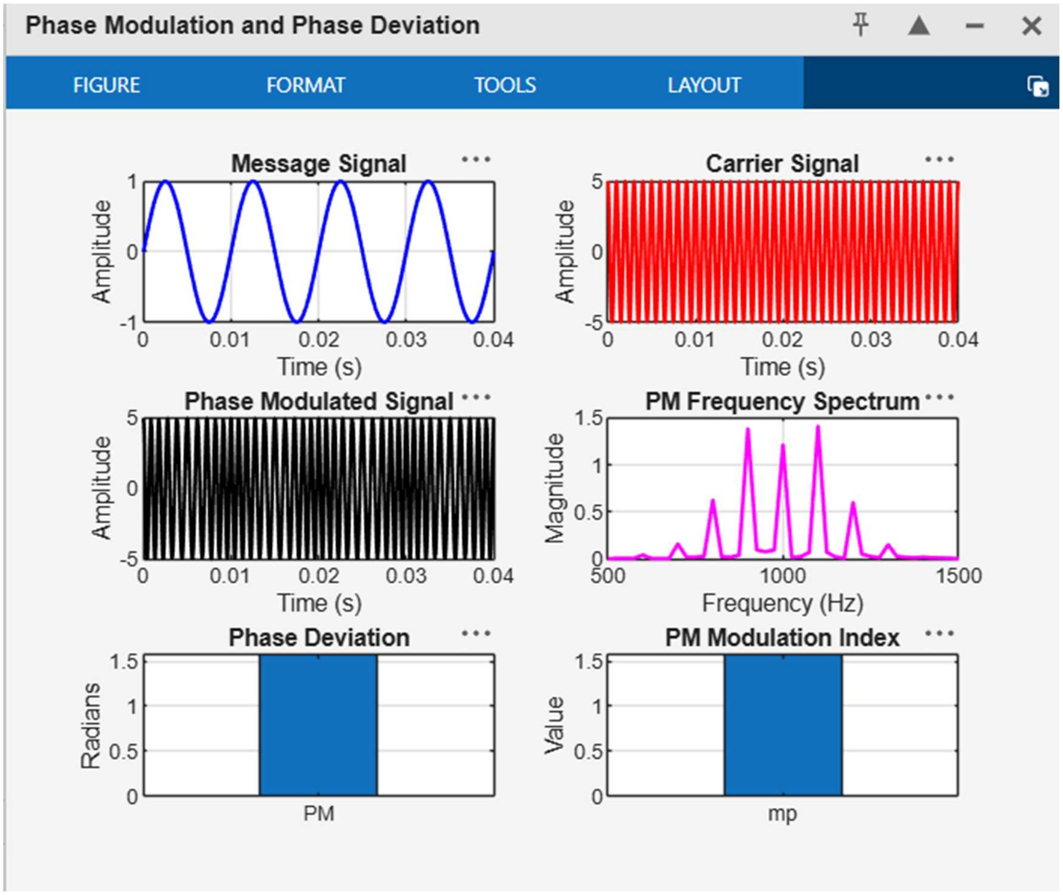
% Carrier Signal
subplot(3,2,2);
plot(t, c, 'r', 'LineWidth', 1.5);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Phase Modulated Signal
subplot(3,2,3);
plot(t, pm, 'k', 'LineWidth', 1.5);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Frequency Spectrum
subplot(3,2,4);
plot(f, PM_FFT, 'm', 'LineWidth', 1.5);
xlim([fc-500 fc+500]);
title('PM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

% Phase Deviation
subplot(3,2,5);
bar(phase_dev);
title('Phase Deviation');
ylabel('Radians');
set(gca,'XTickLabel',{'PM'});
grid on;

% Modulation Index
subplot(3,2,6);
bar(mp);
title('PM Modulation Index');
ylabel('Value');
set(gca,'XTickLabel',{'mp'});
grid on;
```


Phase Modulation and Phase Deviation
FIGURE
FORMAT
TOOLS
LAYOUT
Message Signal
Amplitude
Time (s)
Carrier Signal
Amplitude
Time (s)
Phase Modulated Signal
Amplitude
Time (s)
PM Frequency Spectrum
Magnitude
Frequency (Hz)
Phase Deviation
Radians
PM
PM Modulation Index
Value
mp